# AI Quiz #01 Notes (Corrected)

**Syllabus:** Intro, History, Agents and Environments, Types of Agents.

**Sources:** 01-AI-intro.pptx (Intro and History), 1-intelligent-agent.pdf (Koehn), 2_AI-Agents.pptx (your instructor's agents deck). Search slides are not in the syllabus and are left out. Where Koehn and the instructor deck use different words, both are given. Items marked *Example (not on slides)* or *Worked answer* are explanations added for understanding, not slide content.

# PART 1: INTRODUCTION

## 1.1 AI in Fiction versus AI in Reality

**AI in fiction (film, year, director):**

- Metropolis (1927, Fritz Lang)
- 2001: A Space Odyssey (1968, Stanley Kubrick)
- Star Wars (1977, George Lucas)
- Blade Runner (1982, Ridley Scott)
- Terminator (1984, James Cameron)
- A.I. Artificial Intelligence (2001, Steven Spielberg)
- I, Robot (2004, Alex Proyas)
- Wall-E (2008, Andrew Stanton)
- Ex Machina (2015, Alex Garland)
- Blade Runner 2049 (2017, Denis Villeneuve)

**AI in reality.** Many fantasies from fiction are becoming real:

- self-driving cars
- autonomous vacuum cleaners
- soccer-playing robots
- acrobatic humanoid robots (Boston Dynamics)
- autonomous trading bots
- vocal assistants (Amazon Alexa)
- image and face recognition tools
- world-champion-beating chess players
- world-champion-beating Go players
- AI fighter pilots

## 1.2 Intelligence versus Artificial Intelligence

- **Intelligence.** For thousands of years humans have tried to understand how we think: how can a “handful of matter” perceive, understand, predict and manipulate a world far larger and more complicated than itself? It involves many disciplines: logic, psychology, cognitive science, neuroscience, philosophy, ethics, linguistics.
- **Artificial Intelligence (AI).** Goes further: it attempts not just to understand but also to **build** intelligent entities. It involves all the disciplines above plus mathematics, computer science, engineering, economics, control theory and cybernetics, electronics.

**Short version:** Intelligence studies how minds work. AI tries to build one.

## 1.3 What is Intelligence?

Intelligence has been defined in many ways: the capacity for logic, understanding, self-awareness, learning, emotional knowledge, reasoning, planning, creativity, critical thinking and problem-solving. More generally, it is the ability to **perceive or infer information**, and to **retain it as knowledge** to be applied towards adaptive behaviours within an environment or context.

**Slide example: the simple puzzle.** A visual pattern puzzle whose answer is option B (column-by-column clockwise rotation). The question asked is: what did you do to solve it?

- 1. Read and recognise the figures = **perceive information**.
- 2. Recognise patterns, the problem and candidate solutions = **retain knowledge**.
- 3. Choose the solution = **infer other knowledge**.

## 1.4 What is Artificial Intelligence? The Four Approaches

There is no single definition because of different criteria. Historically there are four approaches along two orthogonal dimensions:

- Dimension 1: thought processes and reasoning **versus** behaviour and action.
- Dimension 2: success measured against human standards **versus** success measured against an ideal concept of intelligence, called **rationality**.

| | **Human standard** | **Rationality** |
|---|---|---|
| **Thought / reasoning** | 1. Thinking Humanly (cognitive modelling) | 3. Thinking Rationally (laws of thought) |
| **Behaviour / action** | 2. Acting Humanly (Turing Test) | 4. Acting Rationally (rational agent) |

The human-centred approaches involve observations and hypotheses about human behaviour. The rationalist approach is a combination of mathematics and engineering. The four groups have both disparaged and helped each other.

### Approach 1: Thinking Humanly (cognitive modelling)

- Problem: how do humans think?
- Idea: develop a theory of the mind, then express the theory as computer programs. Example: Newell and Simon's General Problem Solver (1961).
- Requires scientific theories of brain activity (a cognitive model). This gave the inter-disciplinary field of **Cognitive Science**: it combines computer models from AI with experimental techniques from psychology to construct precise and testable theories of the human mind.
- AI and Cognitive Science are now distinct: AI tries to find an algorithm that performs well on a task; Cognitive Science tries to find a good model of human performance. They still fertilise each other (for example in computer vision).

### Approach 2: Acting Humanly (the Turing Test)

- Problem: when does a system behave intelligently?
- Alan Turing, “Computing Machinery and Intelligence” (1950). An operational test of intelligence, also called **The Imitation Game**.
- Setup: a human, a computer and an interrogator, each in a different room. The interrogator must classify which is the human and which is the machine. If the computer misleads the interrogator and is classified as human, it passes.
- Core idea: “behave intelligently” is treated as equal to “behave humanly”.

**Capabilities needed to pass the Turing Test:**

- Natural language processing: communicate successfully in English (or another language).
- Knowledge representation: store what it knows or hears.
- Automated reasoning: answer questions and draw new conclusions.
- Machine learning: adapt to new circumstances and detect and extrapolate patterns.

**Total Turing Test** (with physical interaction with the interrogator) adds:

- Computer vision: perceive objects. • Computer speech: communicate orally. • Robotics: manipulate objects and move about.

These disciplines make up most of AI. The Turing Test is still relevant in AI, although not fundamental.

**Successes:** in 2014 a chatbot (Eugene Goostman), mimicking the answers of a 13-year-old boy, passed the test. Chatbots are now frequently available and vocal assistants are in common use (Alexa/Amazon, Siri/Apple, Cortana/Microsoft).

**Limitations:**

- Not reproducible, constructive or amenable to mathematical analysis.
- AI researchers devoted little effort to making systems pass it (the slide also asks: do humans always pass it?).
- Should we really emulate humans? Shouldn't we study the underlying principles of intelligence instead?

**Metaphor:** successful flying machines were not built by imitating birds, but by studying engines and aerodynamics.

### Approach 3: Thinking Rationally (the "Laws of Thought")

- Problem: can we capture the laws of thought?
- Aristotle asked what the correct argument and thought processes are. He codified "right thinking" as irrefutable reasoning processes called **syllogisms**. Example: "All men are mortal; Socrates is a man; therefore, Socrates is mortal."
- Leads to Logic and logical inference. The Logicist tradition hopes to build intelligent systems with logic-based inference. Slogan: **"algorithm = logic + control"**. Examples: logic programming, automated-deduction systems; logics: propositional, first-order, modal and description, temporal.

**Two main limitations:**

- It is not easy to state informal knowledge in the formal terms of logic.
- Problems can be undecidable or computationally very hard (NP-hard).

Logical reasoning is still part of many AI fields (problem solving, knowledge representation and reasoning, planning) but does not exhaustively cover AI.

### Approach 4: Acting Rationally (the rational-agent approach)

- Problem: can we make systems "do the right thing"?
- An **agent** is an entity that perceives and acts, and persists over a prolonged time period.
- A **rational agent** acts to achieve the best outcome (maximise goal achievement), or the best *expected* outcome under uncertainty.
- Rational agents need all the skills needed for the Turing Test. Thinking rationally is sometimes part of being a rational agent (for example, planning an action); sometimes it acts without thinking (for example, reflexes).

**Two advantages over earlier approaches:**

- More general than the laws-of-thought approach: correct inference is just one of several mechanisms for achieving rationality.
- More amenable to scientific development than human-emulation approaches: rationality is mathematically well defined and general.

**Remark (limited rationality):** perfect rationality is not feasible in complex environments because the computational demands are too high. It is still a good working hypothesis and starting point for analysis. The practical version is **limited rationality**: acting appropriately when there is not enough time to do all the computation. This course concentrates on rational agents (following the AIMA book).

## 1.5 AI Systems Classification

- **Weak AI vs Strong AI.** Weak AI: is it possible to build systems that *act as if* they were intelligent? Strong AI: is it possible to build systems that *are* intelligent (conscious minds, wills and sentiments)?
- **General AI vs Narrow AI.** General AI copes with any generalised task, much like a human. Narrow AI handles one particular task: it shows intelligence only in a narrow field, performing highly specialised tasks.
- **Symbolic (top-down) vs Connectionist (bottom-up).** Symbolic: symbolic representation of knowledge (logics, ontologies, rule-based systems, declarative architecture); human-understandable models. Connectionist: based on neural networks; knowledge is not symbolic but "encoded" in connections between neurons; concepts are learned from examples; not understandable by humans.

***Example (not on slides):*** *a chess engine is narrow AI: brilliant at chess, useless at anything else.*

## 1.6 AI: State of the Art

**AI is everywhere:** search engines; route planning (maps, traffic); logistics (packages, inventory, airlines); medical diagnosis and machine diagnosis; automated help desks; spam and fraud detection; smarter devices such as cameras; product recommendations; assistants and smart homes.

**What AI systems can currently do:** classify e-mails as spam; predict stock price evolution; understand handwriting; learn to grab a cup; design a molecule with given properties; translate text from Chinese to English; convert voice into text; predict traffic trajectories; automatically write the caption of a figure; drive autonomously; run and jump on two legs; beat a top-gun pilot in a simulated F-16 dogfight.

## 1.7 Main AI Research Venues

- **Journals:** Artificial Intelligence; Computational Intelligence; Journal of Artificial Intelligence Research; IEEE Transactions on Pattern Analysis and Machine Intelligence; IEEE Intelligent Systems.
- **Conferences:** IJCAI (International Joint Conference on AI), AAAI (National Conference on AI), ECAI (European Conference on AI).
- **Professional societies:** AAAI (American Association for Artificial Intelligence), ACM SIGART (Special Interest Group in AI), AISB (Society for AI and Simulation of Behaviour).

# PART 2: HISTORY OF AI

## 2.1 Foundations: Which Fields Contributed to AI

| Field | What it gave AI |
|---|---|
| **Philosophy** | Logic, reasoning, mind as a physical system, foundations of learning, language and rationality |
| **Mathematics** | Formal representation and proof, computation, (un)decidability, (in)tractability, probability |
| **Economics** | Formal theory of rational decisions, game theory |
| **Neuroscience** | Physical substrate for mental activities |
| **Psychology** | Adaptation, phenomena of perception and motor control |
| **Computer Science and Engineering** | Algorithms, data structures, efficient implementations |
| **Control Theory and Cybernetics** | Homeostatic systems, stability, optimal agent design |
| **Linguistics** | Knowledge representation, grammar |

## 2.2 Timeline of AI History

### Era 1: The Gestation of AI (1943–1955)

- 1943: Warren McCulloch and Walter Pitts proposed a model of artificial Boolean neurons that perform computations. First steps toward connectionist computation and learning.
- 1950: Alan Turing, “Computing Machinery and Intelligence”. Introduced the Turing Test; first complete vision of AI.
- 1951: Marvin Minsky and Dean Edmonds (written “Dann Edmonds” on the slide) built the first neural network computer.

### Era 2: The Birth of AI (1956) and the Era of Great Expectations (1952–1969)

- **1956, Dartmouth Workshop** (spelled “Darmouth” on the slide): brought together top minds on automata theory, neural nets and the study of intelligence. Considered the birth of AI as a field.
- **Logic Theorist** (Allen Newell and Herbert Simon): first nonnumerical thinking program; used for theorem proving; proved theorems from Russell and Whitehead's *Principia Mathematica*.
- **General Problem Solver (GPS)** (Newell and Simon): handled a limited number of logical puzzles; imitated human problem solving with a strategy of addressing subgoals. Idea: any system, human

or machine, exhibiting intelligence must operate by manipulating data structures composed of symbols.

- **John McCarthy:** invented LISP (and time-sharing); proposed the logic-oriented Advice Taker, decoupling knowledge and reasoning.
- **Marvin Minsky:** microworlds, problems in a limited domain that appear to require intelligence (blocks world, geometric problems).
- Winograd, Cowan and others: early work on neural networks.

### Era 3: Collapse of AI Research (1966–1973)

- Progress was slower than expected; enthusiastic predictions turned unrealistic.
- Some systems lacked scalability: computational intractability due to combinatorial explosion in search.
- Fundamental limitations on the techniques and representations used.
- Minsky and Papert (1969): important limitations of neural networks.

### Era 4: Revival via Knowledge-Based Systems (1969–1970 on the slide)

- Shift from general-purpose to domain-specific systems: narrow domains, exploiting domain-specific knowledge.
- **DENDRAL** (Buchanan et al., 1969): inferred molecular structure from mass-spectrometer information.
- **Expert systems** applied to areas of human expertise. **MYCIN** (Feigenbaum et al.): diagnosed blood infections using about 450 domain-specific rules from experts and textbooks, plus a calculus for uncertainty.
- Several advances in natural language processing by incorporating domain knowledge.

### Era 5: AI Becomes an Industry (1980–present)

- Commercial expert system **R1** at DEC (McDermott, 1982) helped configure orders for computer systems, saving about $40 million per year.
- A period of national and industry investment followed, then an expert-systems industry bust: the **“AI Winter”**.

### Era 6: The Return of Neural Networks (1986–present)

- The **back-propagation** learning algorithm was (re)invented and applied to many learning problems in computer science and psychology.
- Revival of connectionist models, versus symbolic or logicist approaches.

### Era 7: AI Adopts the Scientific Method (1987–present)

- A “gentle revolution” in AI content and methodology: build on existing theories rather than propose brand-new ones; base claims on rigorous theorems or hard experimental evidence rather than intuition; show relevance to real-world applications rather than toy examples.
- Hypotheses must be subjected to rigorous empirical experiment, and results analysed statistically for their importance. Result: a general increase in technical depth.
- Resurgence of probability and focus on uncertainty: hidden Markov models (speech and handwriting recognition); Bayesian networks for rigorous reasoning with uncertainty. Neural networks benefited from statistics, pattern recognition and machine learning, leading to data mining.
- Similar gentle revolutions occurred in robotics, computer vision and knowledge representation.

### Era 8: The Emergence of Intelligent Agents (1995–present)

- Renewed interest in the “whole agent” problem: how does an agent act and behave embedded in real environments with continuous sensory inputs? Examples: AI in the internet domain (“-bots”), decision support systems, robotic agents, natural language.
- Need for interaction between sensing and reasoning, so reasoning and planning systems must handle uncertainty.
- AI was forced into much closer contact with other fields, such as control theory and economics.

**Era 9: Availability of Very Large Data Sets (2001–present)**

- Big data and massive computing power (for example GPUs) enabled deep networks to be properly trained.
- Emphasis moved from algorithms to data. Learning methods, rather than hand-coded knowledge engineering, express the knowledge a system needs.
- Large variety of applications (speech and image recognition, spam filtering, robotics, machine translation, autonomous vehicles, game playing), now deeply embedded in the infrastructure of every industry.

**Era 10: The Deep-Learning Tsunami (2015–present)**

- C. Manning: 2015 seems the year when the full force of the deep-learning "tsunami" hit the major NLP conferences. Earlier successes were in image classification and speech.
- LeCun, Hinton and Bengio (Turing Award winners) expect important developments in text and video understanding, machine translation and question answering.
- **AlphaGo** (Google): deep-learning software defeated a human professional Go player for the first time (Nature, 28 January 2016). In March 2016 it beat Lee Sedol (the slide's wording here is garbled; it is AlphaGo that won).

**Memory chain:** 1943 neurons → 1950 Turing → 1956 Dartmouth (birth) → 1966–73 collapse → 1969 expert systems → 1980 industry, then AI Winter → 1986 back-propagation → 1987 scientific method → 1995 agents → 2001 big data → 2015 deep learning.

# PART 3: AGENTS AND ENVIRONMENTS

## 3.1 What is an Agent?

**Agent (instructor deck):** anything that can be viewed as perceiving its environment through **sensors** and acting upon that environment through **actuators**.

**Intelligent agent (Koehn):** perceives its environment via sensors and acts *rationally* upon it with its **effectors**. (Effectors and actuators mean the same thing.) Agents include humans, robots, softbots, thermostats, etc.

- **Percept:** what the agent senses at one moment. **Percept sequence / history:** everything sensed so far.
- **Action:** what the agent does through its actuators.

**Humans as an example (Koehn):**

- Sensors: eyes (vision), ears (hearing), skin (touch), tongue (gustation), nose (olfaction), neuromuscular system (proprioception).
- Percepts: at the lowest level, electrical signals from these sensors; after preprocessing, objects in the visual field (location, textures, colours) and auditory streams (pitch, loudness, direction).
- Effectors: limbs, digits, eyes, tongue. Actions: lift a finger, turn left, walk, run, carry an object.
- Percepts and actions need to be carefully defined, possibly at different levels of abstraction.

## 3.2 Agent Function, Agent Program, Architecture

- **Agent function:** maps percept histories to actions, **f : P* → A**. An agent is completely specified by its agent function.
- **Agent program:** runs on the physical architecture to produce f.
- **agent = architecture + program**
- One agent function (or a small equivalence class of them) is rational. Aim: find a way to implement the rational agent function concisely.

## 3.3 Example: The Vacuum-Cleaner World

Two squares, A and B. Percepts: location and contents, e.g. [A, Dirty]. Actions: Left, Right, Suck, NoOp.

| Percept sequence | Action |
|---|---|
| **[A, Clean]** | Right |
| **[A, Dirty]** | Suck |
| **[B, Clean]** | Left |
| **[B, Dirty]** | Suck |
| **[A, Clean], [A, Clean]** | Right |
| **[A, Clean], [A, Dirty]** | Suck |
| **...** | ... |

**Function version (Koehn):** input location, status. If status = Dirty return Suck; if location = A return Right; if location = B return Left.

The program is tiny, while the table grows with every possible percept sequence. Questions on the slide: *What is the right function? Can it be implemented in a small agent program?*

## 3.4 Rationality

- **Performance measure:** an objective criterion for the success of an agent's behaviour. A fixed performance measure evaluates the *environment sequence*.
- Vacuum-cleaner measures: amount of dirt cleaned up, time taken, electricity consumed, noise generated; one point per square cleaned in time T; one point per clean square per time step minus one per move; penalise for more than k dirty squares.
- Other types of performance measure (Koehn): false alarm (false positive) rate, false dismissal (false negative) rate, speed, resources required, impact on environment.
- **Rational agent:** for each possible percept sequence, selects the action expected to maximise its performance measure, given the evidence from the percept sequence and whatever built-in (and acquired) knowledge the agent has.
- Rationality includes information gathering, not “rational ignorance”: if you don't know something, find out.

**Key quiz idea:**

- Rational ≠ omniscient: percepts may not supply all relevant information.
- Rational ≠ clairvoyant: action outcomes may not be as expected.
- Hence rational ≠ successful.
- Rational implies exploration, learning and autonomy. Exploration: agents can act to modify future percepts to obtain useful information.

***Example (not on slides):*** *you check both ways and cross an empty street; a cargo door falls from an aircraft and hits you. Crossing was rational (best expected result from your percepts) but not successful, because nobody could have predicted the falling door.*

## 3.5 Autonomy

- Autonomous = the agent's behaviour is determined by its own experience (with the ability to learn and adapt).
- If its behaviour is guided only by its designer's built-in knowledge, it is not autonomous.
- To survive, an agent needs enough built-in knowledge to survive **and** the ability to learn.

***Example (not on slides):*** *a thermostat following only a programmed rule is not autonomous; a robot vacuum that learns your house layout from experience is more autonomous.*

## 3.6 Some Properties of Agents (Koehn)

- A **discrete** agent receives percepts one at a time and maps this percept sequence to a sequence of discrete actions.
- Other possible properties: autonomous; reactive to the environment; pro-active (goal-directed); interacts with other agents via the environment.

## 3.7 Example: Self-Driving Car (Koehn)

- Percepts: video, lidar, speedometer, odometer, engine sensors, keyboard input, microphone, GPS.
- Actions: steer, accelerate, brake, horn, speak/display.
- Goals: maintain safety, reach destination, maximise profits (fuel, tire wear), obey laws, speed, passenger comfort.
- Environment: (U.S.) urban streets, freeways, traffic, pedestrians, weather, customers.
- Different aspects of driving may require different types of agent programs.

## 3.8 PEAS: Specifying the Task Environment

**P**erformance measure, **E**nvironment, **A**ctuators, **S**ensors. You must first specify the setting for intelligent agent design.

| Agent | Performance | Environment | Actuators | Sensors |
|---|---|---|---|---|
| **Automated taxi driver** | Safe, fast, legal, comfortable trip, maximise profits | Roads, other traffic, pedestrians, customers | Steering wheel, accelerator, brake, signal, horn | Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard |
| **Medical diagnosis system** | Healthy patient, minimise costs, lawsuits | Patient, hospital, staff | Screen display (questions, tests, diagnoses, treatments, referrals) | Keyboard (symptoms, findings, patient's answers) |
| **Part-picking robot** | Percentage of parts in correct bins | Conveyor belt with parts, bins | Jointed arm and hand | Camera, joint angle sensors |
| **Interactive English tutor** | Maximise student's score on test | Set of students | Screen display (exercises, suggestions, corrections) | Keyboard |

**Class exercises on the slides** (the slides give no answers; the answers below are **worked answers, not from the slides**). The slide for the robot soccer player asks for PEAS *and all environment types*.

| | Internet book-shopping agent | Robot soccer player |
|---|---|---|
| **P** | Right book found, low price, fast delivery, trustworthy seller, within budget | Goals scored, goals conceded, matches won, ball possession |
| **E** | Internet, online bookstores, sellers, payment systems, the user | Field, ball, teammates, opponents, referee |
| **A** | Display results/recommendations, fill forms, click links, place the order | Legs and feet (kick, walk, run), turning motors, radio/speaker to talk to teammates |
| **S** | User's keyboard/mouse input, web page content (HTML), search results, price and stock data | Camera, distance sensors, gyroscope, joint sensors, radio |
| **Environment types** | Partially observable, deterministic, sequential, semidynamic, discrete, single agent | Partially observable, stochastic, sequential, dynamic, continuous, multiagent (cooperative with teammates, competitive with opponents) |

## 3.9 Environment Types (Properties of Environments)

Terms in brackets are Koehn's wording for the same property.

### 1. Fully observable vs partially observable (accessible vs inaccessible)

- Fully observable: the agent's sensors give it access to the complete state of the environment at each point in time. Convenient, since the agent is freed from keeping track of changes in the environment.
- Partially observable: sensors give only part of the state.

- Examples: checkers is fully observable (you see how many pieces each side has and where). Poker is partially observable (you don't know the opponent's cards and decide from the cards played).

### 2. Deterministic vs stochastic (deterministic vs nondeterministic)

- Deterministic: the next state is completely determined by the current state and the action executed by the agent. In an accessible and deterministic environment the agent need not deal with uncertainty.
- **Strategic:** deterministic except for the actions of other agents.
- Example: moving a chess pawn from A2 to A3 always works, with no uncertainty in the outcome. Backgammon is nondeterministic because of the dice.

### 3. Episodic vs sequential

- Episodic: experience is divided into atomic episodes (the agent perceives and then performs a single action); the choice of action in each episode depends only on the episode itself. Subsequent episodes do not depend on previous ones, so the agent does not need to plan ahead.
- Sequential: the current decision could affect all future decisions. Episodic environments are much simpler because the agent need not think ahead.
- Examples: spotting defective parts on an assembly line is episodic (each decision uses the current part; it does not affect whether the next part is defective). Chess and taxi driving are sequential: short-term actions have long-term consequences.

### 4. Static vs dynamic (and semidynamic)

- Static: the environment is unchanged while the agent deliberates. Easy to deal with: the agent need not keep looking at the world or worry about the passage of time. Example: crossword puzzle.
- Dynamic: continuously asking the agent what it wants to do; if it hasn't decided yet, that counts as deciding to do nothing. Example: taxi driving (other cars and the taxi keep moving).
- Semidynamic: the environment itself does not change with time, but the agent's performance score does. Example: chess played with a clock.

### 5. Discrete vs continuous

- Discrete: a limited (finite) number of distinct, clearly defined percepts and actions. Example: checkers (finite board positions and moves).
- Continuous: values sweep smoothly through a range. Example: a patient's temperature and blood pressure.
- The distinction applies to the state of the environment, to the way time is handled, and to the percepts and actions. Taxi driving is continuous-state and continuous-time (speed and location vary smoothly), and its actions (steering angles) are continuous too.

### 6. Single agent vs multiagent

- Single agent: an agent operating by itself (solving a crossword puzzle). Chess is a two-agent environment.
- Multiagent can be **competitive** (chess) or **cooperative** (taxis avoiding collisions).

### 7. With or without intelligent adversaries (Koehn)

- If the environment contains intelligent, adversarial agents, the agent must be concerned about strategic, game-theoretic aspects.
- Most engineering environments have no rational adversaries, whereas most social and economic systems get their complexity from the interactions of (more or less) rational agents.

### 8. Uncertain environment

- Uncertain if it is **not fully observable OR not deterministic**.

**Summary table (instructor deck):**

| Property | Chess with clock | Chess without clock | Taxi driving |
|---|---|---|---|
| **Fully observable** | Yes | Yes | No |
| **Deterministic** | Strategic | Strategic | No |

| Property | Chess with clock | Chess without clock | Taxi driving |
|---|---|---|---|
| **Episodic** | No | No | No |
| **Static** | Semi | Yes | No |
| **Discrete** | Yes | Yes | No |
| **Single agent** | No | No | No |

- The environment type largely determines the agent design.
- The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous and multi-agent.
- Koehn: the most challenging environments are inaccessible, nondeterministic, nonepisodic, dynamic and continuous, and contain intelligent adversaries.

**Koehn's classification exercise** (filled in on the final slides):

| Environment | Accessible | Deterministic | Episodic | Static | Discrete |
|---|---|---|---|---|---|
| **Image classification** | yes | yes | yes | yes | no |
| **Solitaire** | no | yes | no | yes | yes |
| **Backgammon** | yes | no | no | yes | yes |
| **Customer support chat** | no | no | no | no | yes |
| **Taxi driving** | no | no | no | no | no |

**Reasoning (explanations are mine, answers are the slide's):** Solitaire is not accessible because cards are face-down, but it is deterministic and static, and earlier moves affect later ones (sequential). Backgammon: the whole board is visible, but dice make it nondeterministic. Customer support chat: you can't see what the customer thinks or will say next, it depends on conversation history, and the customer keeps typing while you decide; messages are discrete units of text.

# PART 4: TYPES OF AGENTS

## 4.1 Table-Lookup (Table-Driven) Agent

Uses a percept-sequence/action table in memory to find the next action; implemented by a (large) lookup table that maps every possible perceived state to the optimal action.

**Drawbacks:**

- Huge table: too big to generate and to store (chess has about $10^{120}$ states).
- Takes a long time to build the table.
- No autonomy; even with learning, it takes a long time to learn the table entries.
- No knowledge of non-perceptual parts of the current state.
- Not adaptive: if the environment changes, the entire table must be updated.
- Looping: can't make actions conditional on previous actions/states.

## 4.2 The Five Basic Agent Types (instructor deck)

In order of increasing generality: **1. Simple reflex, 2. Model-based reflex, 3. Goal-based, 4. Utility-based, 5. Learning** agents.

Koehn lists them as: table-driven, simple reflex, agents with memory, agents with goals, utility-based. “Agents with memory” = model-based reflex agents.

## 4.3 Simple Reflex Agents

- Based on condition-action rules, implemented with an appropriate production system (rule-based reasoning from percepts to action; each rule handles a collection of perceived states).
- Stateless: no memory of past world states.
- Architecture: Sensors → “What the world is like now” → (with condition-action rules) → “What action I should do now” → Effectors.

- Example: the vacuum-cleaner function above (if dirty, suck; else move).

**Problems:** still usually too big to generate and store; still no knowledge of non-perceptual parts of the state; still not adaptive (rules must be updated if the environment changes); still can't make actions conditional on previous state.

> ***Example (not on slides):*** *automatic doors, or a car that brakes when its camera sees brake lights ahead.*

## 4.4 Model-Based Reflex Agents (Agents with Memory)

- Have an internal **state** used to keep track of past states of the world (to incorporate past percepts).
- Needed because sensors do not usually give the entire state of the world at each input, so perception of the environment is captured over time.
- State is used to encode different world states that generate the same percept.
- Architecture adds three inputs to “What the world is like now”: **State**, **How the world evolves**, **What my actions do**. Condition-action rules then choose the action.

> ***Example (not on slides):*** *a car moves into your blind spot; you remember it is there so you don't change lanes. A simple reflex agent would see an empty mirror and change lanes.*

**Example architecture: Rodney Brooks' Subsumption Architecture.**

- Main idea: build complex, intelligent robots by decomposing behaviours into a hierarchy of skills.
- Each skill is a complete percept-action cycle for one very specific task (avoiding contact, wandering, exploring, recognising doorways).
- Behaviours are loosely coupled, asynchronous interactions.

## 4.5 Goal-Based Agents

- In addition to state information, they have **goal information** describing desirable situations; they choose actions to achieve a given or computed goal. A goal is a description of a desirable situation.
- Keeping track of the current state is often not enough; goals are needed to decide which situations are good.
- **Deliberative instead of reactive:** they take future events into consideration and may consider long sequences of possible actions to decide if the goal is achieved (“what will happen if I do...?”). This leads to search and planning, covered in later chapters.
- Architecture: model-based parts plus “What it will be like if I do action A”, which is compared with **Goals** to give “What action I should do now”.

> ***Example (not on slides):*** *a taxi at a junction with the goal “reach the airport” works out which turn leads there. Change the goal and behaviour changes without rewriting rules.*

## 4.6 Utility-Based Agents

- A goal is a binary choice: happy or unhappy. Utility gives a performance measure of “degree of happiness”, based on classic axiomatic utility theory, in order to act rationally.
- **Utility function U : State → Real numbers**, a measure of success or happiness at a given state.
- Allows decisions comparing (a) conflicting goals, and (b) likelihood of success versus importance of a goal, when achievement is uncertain.
- Architecture: goal-based parts, but “Utility” feeds “How happy I will be in such a state”, which gives “What action I should do now”.

> ***Example (not on slides):*** *a taxi has three routes to the airport: very fast but risky (utility 6), a bit slower but safe and cheap (utility 9), slow and expensive (utility 3). A goal-based agent sees all three reach the goal; a utility-based agent picks the one with utility 9.*

## 4.7 Learning Agents

The instructor deck gives only a diagram for this type. The diagram shows a Performance standard, Critic, Learning element, Problem generator and Performance element, with arrows labelled feedback, changes, knowledge and learning goals. The explanation below follows that diagram (it matches the standard AIMA textbook; check it against your lecture).

- A learning agent can start in an unknown environment and become more competent than its initial knowledge alone allows, making it truly autonomous.
- **Performance element:** selects external actions (it is the whole agent of the earlier types).
- **Learning element:** improves the performance element, using feedback from the critic.
- **Critic:** tells the learning element how well the agent is doing, judged against a fixed *performance standard*.
- **Problem generator:** suggests exploratory actions leading to new, informative experiences.

***Example (not on slides):*** *a chess program loses; the critic says the early loss of the queen was the cause; the learning element adjusts how it values queens; the problem generator tries an unusual opening to see if it works better.*

### 4.8 Comparing the Types

- Simple reflex: current percept only; no memory, no goals.
- Model-based reflex: reflex + internal state/model of the world.
- Goal-based: model + goals; thinks ahead.
- Utility-based: model + utility function; handles trade-offs and uncertainty.
- Learning: can be any of the above, plus the ability to improve from experience.

**Koehn's summary slide:**

- An agent perceives and acts in an environment, has an architecture, and is implemented by an agent program.
- An ideal agent always chooses the action that maximises its expected performance, given its percept sequence so far.
- An autonomous agent uses its own experience rather than the designer's built-in knowledge.
- An agent program maps percept to action and updates its internal state: a reflex agent responds immediately to percepts; a goal-based agent acts to achieve its goals; a utility-based agent maximises its own utility function.
- Representing knowledge is important for successful agent design.

## FINAL REVISION CHECKLIST

1. Define intelligence and AI, and say how they differ.
2. Name the four approaches to AI with an example each (GPS; Imitation Game; Aristotle's syllogisms; best expected outcome).
3. Capabilities for the Turing Test, plus the three extra for the Total Turing Test.
4. Two limitations of the Turing Test and two of the laws-of-thought approach.
5. Weak vs strong AI, general vs narrow AI, symbolic vs connectionist AI.
6. Timeline 1943 to 2015 with key names (McCulloch and Pitts, Turing, Dartmouth, Newell and Simon, McCarthy, Minsky, DENDRAL, MYCIN, R1, back-propagation, AlphaGo).
7. Define agent, percept, sensor, actuator, agent function, agent program; agent = architecture + program.
8. Explain rational ≠ omniscient and rational ≠ successful, with an example.
9. Define autonomy.
10. Write PEAS for any given agent.
11. Define each environment property with an example (observable, deterministic/stochastic/strategic, episodic/sequential, static/dynamic/semidynamic, discrete/continuous, single/multiagent).
12. Classify chess (with and without a clock) and taxi driving on all properties.
13. List the drawbacks of table-driven agents.
14. Describe all five agent types and what each adds over the previous one, with an example.